

Factorial benchmark design

Orthogonal factors expose adaptation × retrieval interactions

Held-out test item
Q · source context · reference answer



01 Factor levels

MODEL
Gemma 4
MedGemma 1.5

STATE
 W_0 Base
 $W_0 + \text{BA}$ QLoRA

RETRIEVAL
∅ no_rag
≠ hybrid
≠ HyDE



02 Model × adaptation matrix

	BASE	GROUNDING QLoRA
Gemma 4	W_0	$W_0 + \text{BA}$
MedGemma 1.5	W_0	$W_0 + \text{BA}$



$\underbrace{2}_{\text{models}} \times \underbrace{2}_{\text{states}} \times \underbrace{3}_{\text{retrieval modes}} = 12 \text{ conditions}$

Paired inference
greedy decoding

Provenance
condition + qa_id

RAGAS
[F AR AC SS]